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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Artificial Intelligence and Data Science

Academic Year: 2024- 2025 (ODD Semester)

Active Learning Best practices: Flipped Classroom – Locking Protocols - Two phase locking protocol

Degree, Semester & Branch: III Sem. B.Tech. Artificial Intelligence and Data Science – 'A'

Course Code & Title: - AD3391 – Database Design and Management

Name of the Faculty member: Mrs. G. Kavitha, AP/AI & DS

Theme of discussion: Flipped Classroom

Topics Covered: Unit- IV - Locking Protocols - Two phase locking protocol

Date and Time: 25.10.2024 & 10.50 A.M to 11.35 A.M

Active Learning Best practices: Flipped Classroom – Locking Protocols - Two phase locking protocol

Unit / Topic: Unit – IV / Locking Protocols - Two phase locking protocol

Course Outcome: CO4 - Illustrate the serializability of any non-serial schedule using concurrency control techniques.

Topic Learning Outcome: TLO37

Activity Chosen: Flipped Classroom

Justification:

- The concept of two phase locking protocol lecture was posted into the student's group, to identify the learning level of students about the locking protocol activity was given. It is an important topic. The question was asked in most of the university questions.

Time Allotted for the Activity: 30 Minutes

Details of the Implementation:

The flipped classroom activity was implemented as a teaching strategy to enhance student engagement, foster deeper understanding, and promote active learning. In a traditional classroom, students typically receive lectures during class time and do homework or assignments outside of the class. However, in a flipped classroom, this model is reversed - students watch pre-recorded lectures or read material at home and on the day of activity ask the students to explain about the concept. This report summarizes the implementation and outcomes of the flipped classroom activity.

Locking Protocols - Two phase locking protocol:

In a database management system (DBMS), lock-based concurrency control is used to control the access of multiple transactions to the same data item. This protocol helps to maintain data consistency and integrity across multiple users.

In the protocol, transactions gain locks on data items to control their access and prevent conflicts between concurrent transactions.

What is a Lock?

- A lock is a mechanism to control concurrent access to a data item.
- Data items can be locked in two modes:
  1. exclusive (X) mode. Data item can be both read as well as written. X-lock is requested using lock-X instruction.
  2. shared (S) mode. Data item can only be read. S-lock is requested using lock-S instruction.
- Lock requests are made to concurrency-control manager. Transaction can proceed only after request is granted.

	S	X
S	true	false
X	false	false

- A transaction may be granted a lock on an item if the requested lock is compatible with locks already held on the item by other transactions.
- Any number of transactions can hold shared locks on an item, but if any transaction holds an exclusive on the item no other transaction may hold any lock on the item.
- If a lock cannot be granted, the requesting transaction is made to wait till all incompatible locks held by other transactions have been released. The lock is then granted.
- A locking protocol is a set of rules followed by all transactions while requesting and releasing locks. Locking protocols restrict the set of possible schedules.

Two-Phase Locking Protocol

- This is a protocol which ensures conflict-serializable schedules.
- Phase 1: Growing Phase
  - transaction may obtain locks
  - transaction may not release locks
- Phase 2: Shrinking Phase
  - transaction may release locks
  - transaction may not obtain locks
- The protocol assures serializability. It can be proved that the transactions can be serialized in the order of their lock points (i.e. the point where a transaction acquired its final lock).

- Two-phase locking does not ensure freedom from deadlocks.
- Cascading roll-back is possible under two-phase locking. To avoid this, follow a modified protocol called strict two-phase locking. Here a transaction must hold all its exclusive locks till it commits/aborts.
- Rigorous two-phase locking is even stricter: here all locks are held till commit/abort. In this protocol transactions can be serialized in the order in which they commit.
- There can be conflict serializable schedules that cannot be obtained if two-phase locking is used.
- However, in the absence of extra information (e.g., ordering of access to data), two-phase locking is needed for conflict serializability in the following sense: Given a transaction Ti that does not follow two-phase locking, we can find a transaction Tj that uses two-phase locking, and a schedule for Ti and Tj that is not conflict serializable.

CO – PO / PSO Mapping

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	2	2	2	2	2	2	2	3	2	3

PO/PSO Mapped

Innovative Practice	PO1	PO2	PO3	PO4	PO9	PO10
	3	2	2	2	2	2
Justification for correlation	Students can be able to apply principles of locking protocols to solve real time problems.	Students can be able to analyze schedules to determine their impact on transaction consistency	Students can be able to explore and implement concurrency control mechanisms in database systems.	Students can be able to align the principles and phases of two phase locking protocol with process efficiency, conflict minimization and data integrity.	Students can be able to collaborate effectively in teams to implement and troubleshoot two-phase locking mechanisms	Students can be able to communicate lock concept and two phase locking mechanisms to diverse stakeholders, including technical and non-technical stakeholders.

Innovative Practice	PO11	PO12	PSO1	PSO2	PSO3
	2	2	3	2	3
Justification for correlation	Students can be able to manage projects that involve implementing concurrency control mechanisms.	Students can be able to identify the gaps in transaction management and keep abreast of new concurrency control methods and their applications.	Students can be to apply locking protocol concepts and techniques to solve a variety of engineering problems.	Students can be able to apply locking protocol capabilities in a single database system to support real-time analytics and transactional workloads simultaneously.	Students can be able to apply locking protocols for forecasting future trends.

Glimpses:

Reflective Critique:

- Preparing and Sharing Content: Video lectures and reading materials were created covering the relevant topics. These materials were shared with the students through the learning management system or other suitable platforms well in advance of the corresponding class sessions.
- Class Sessions: During the class sessions, students were asked to discuss and apply the concepts covered in the pre-recorded lectures. Facilitators were present to guide and support the students' learning process.
- Active Learning Activities: Students are asked to explain about the concept in front of the classroom, and peer teaching sessions were conducted.
- Feedback Collection: Students were asked to provide feedback on the flipped classroom experience, highlighting the aspects they found beneficial and suggesting potential improvements.

Observations:

The implementation of the flipped classroom activity was successful in enhancing student engagement, promoting deeper understanding, and creating a more interactive learning environment. By using this approach, students were encouraged to take ownership of their learning process and actively participate in class activities. However, continuous refinement based on student feedback and ongoing assessment of the approach will be essential to optimize its effectiveness in future implementations.

Benefit of the practice:

- The students understand well about use of two phase locking in concurrency control.
- The students can able to interact with the faculty member and their classmates during the class was effective.
- It helps to improve the self-learning of a student.

- With the help of this activity, each student can get clear idea about the use of two phase locking in concurrency control.

Students Response:

- Bright students were actively participated to prepare the topic wise questions and concept notes.
- Slow learners expect to assist the topic towards searching online and also understanding the questions
- Most of the students were enjoyed the session.
- Students response ensures the students could improve listening, communication, creativityand problem-solving skills.

30/10/…

Signature of the Faculty Member

30/10

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Academic Year: 2024- 2025 (ODD Semester)

Active Learning Best practices: Flipped Classroom – Locking Protocols - Two phase locking protocol

Degree, Semester & Branch: III Sem. B.Tech. Artificial Intelligence and Data Science – 'A'

Course Code & Title: - AD3391 – Database Design and Management

Name of the Faculty member: Mrs. G. Kavitha, AP/AI & DS

Theme of discussion: Flipped Classroom

Topics Covered: Unit- IV - Locking Protocols - Two phase locking protocol

Date and Time: 25.10.2024 & 10.50 A.M to 11.35 A.M

Feedback Questions:

1. Did the plan align with your expectations?

   Yes No

2. How effectively was the innovative practice plan implemented?

   Excellent Good Satisfactory

3. How well did you understand the objectives and purpose of the innovative practice plan?

   1 2 3 4 5

4. Were the necessary resources and support provided for successful implementation?

   Yes No

5. Did you notice any specific benefits or changes resulting from the plan?

   Yes No

6. How engaged were you throughout the practice plan?

   1 2 3 4 5

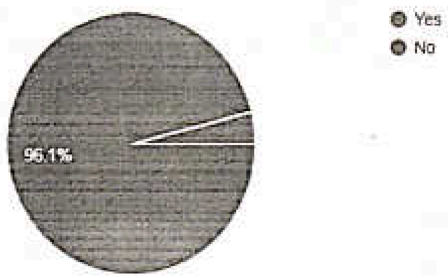
7. Did the plan provide opportunities for active participation and collaboration?

   Yes No

Feedback Analysis:

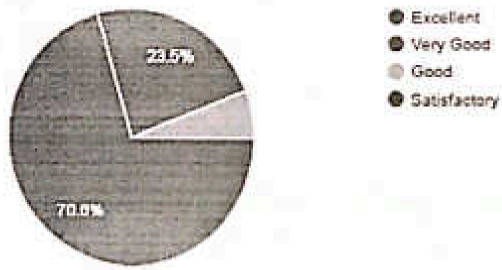
Did the plan align with your expectations? Copy

51 responses



How effectively was the innovative practice plan implemented? Copy

51 responses



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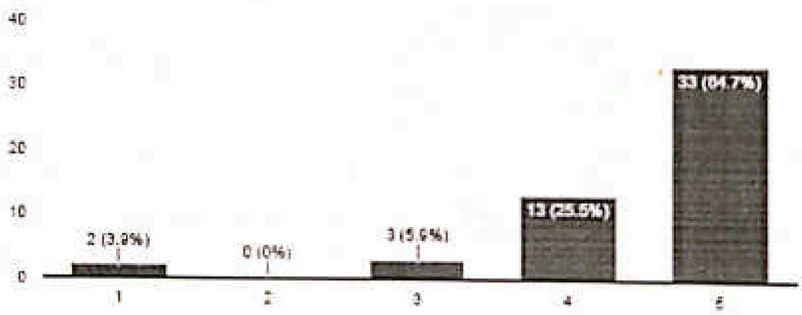

Were the necessary resources and support provided for successful implementation? Copy

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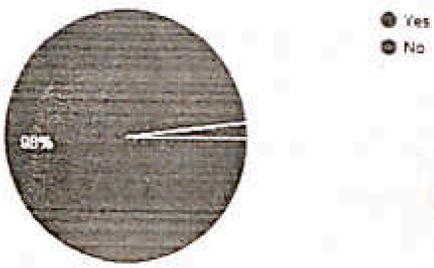

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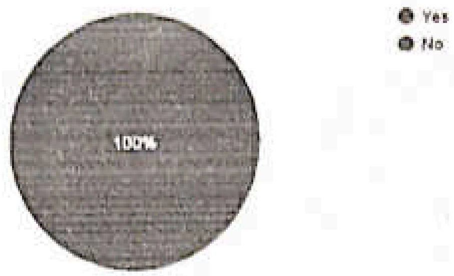

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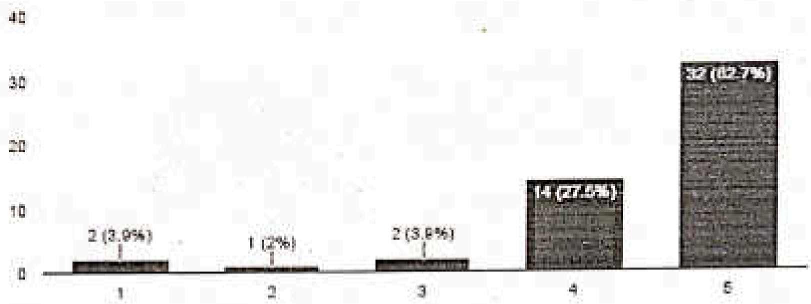

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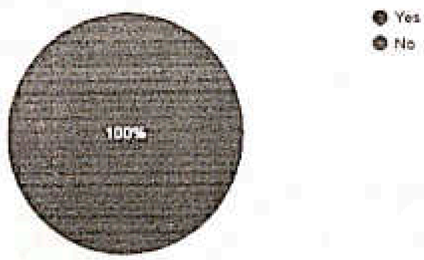

20/10/24

Signature of the Faculty Member

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Active Learning Best practices: Flipped Classroom – Locking Protocols - Two phase locking protocol

Degree, Semester & Branch: III Sem. B.Tech. Artificial Intelligence and Data Science – 'B'
Course Code & Title: AD3391 – Database Design and Management
Name of the Faculty member: Mrs. G. Kavitha, AP/AI & DS

Theme of discussion: Flipped Classroom
Topics Covered: Unit- IV - Locking Protocols - Two phase locking protocol
Date and Time: 28.10.2024 & 01.55 P.M to 02.40 P.M

Active Learning Best practices: Flipped Classroom – Locking Protocols - Two phase locking protocol

Unit / Topic: Unit – IV / Locking Protocols - Two phase locking protocol
Course Outcome: CO4 - Illustrate the serializability of any non-serial schedule using concurrency control techniques.
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4/4/24   4/7

Signature of the Faculty Member   **HoD**

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Degree, Semester & Branch: III Sem. B.Tech. Artificial Intelligence and Data Science – 'B'

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Name of the Faculty member: Mrs. G. Kavitha, AP/AI & DS

Theme of discussion: Flipped Classroom

Topics Covered: Unit- IV - Locking Protocols - Two phase locking protocol

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   Yes No

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4. Were the necessary resources and support provided for successful implementation?

   Yes No

5. Did you notice any specific benefits or changes resulting from the plan?

   Yes No

6. How engaged were you throughout the practice plan?

   1 2 3 4 5

7. Did the plan provide opportunities for active participation and collaboration?

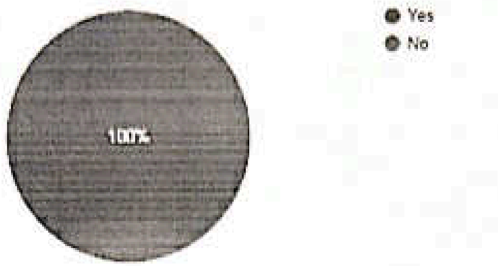
   Yes No

Feedback Analysis:

Did the plan align with your expectations?

Copy

49 responses



How effectively was the innovative practice plan implemented?

Copy

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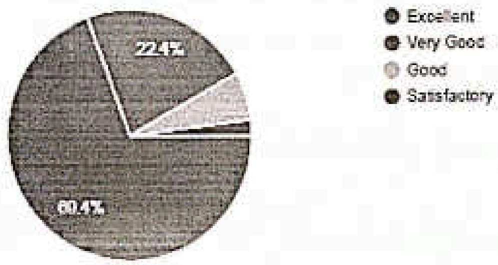

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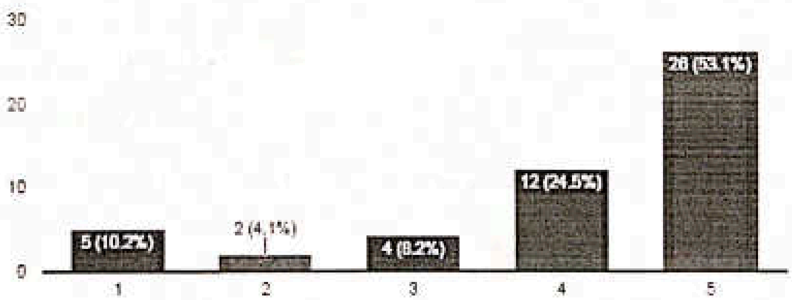

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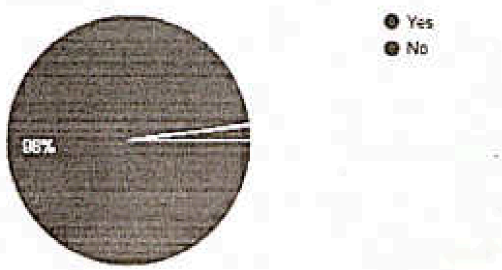

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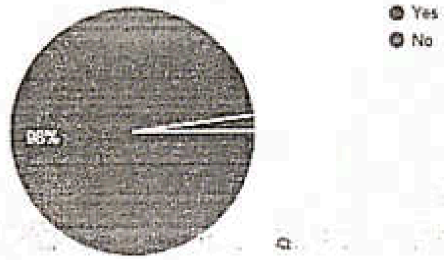

How engaged were you throughout the practice plan?

Copy

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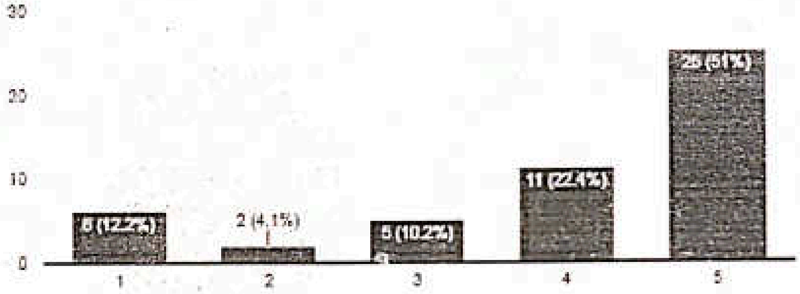

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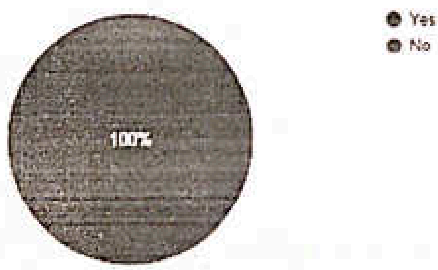

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Signature of the Faculty Member

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